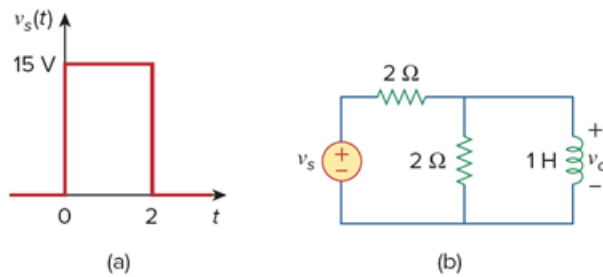


Problem

If the rectangular pulse in Fig. 18.46(a) is applied to the circuit in Fig. 18.46(b), find v_o at $t = 1$ s.

Figure 18.46



Step-by-step solution

Step 1 of 4

Refer to Figure 18.46(b) in the textbook.

Using source transformation, convert the voltage source into current source.

$$I_s = \frac{V_s}{2}$$

Draw the modified circuit in frequency domain.

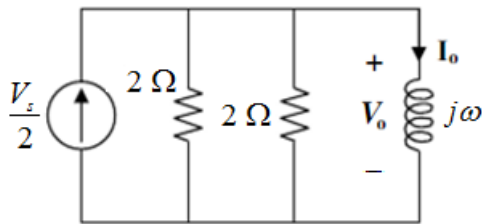


Figure 1

Step 2 of 4

In Figure 1, the resistors are connected in parallel. Therefore, their equivalent resistance is,

$$\begin{aligned} R_{eq} &= (2) \parallel (2) \\ &= \frac{2 \times 2}{2 + 2} \\ &= 1 \Omega \end{aligned}$$

Draw the modified circuit of Figure 1.

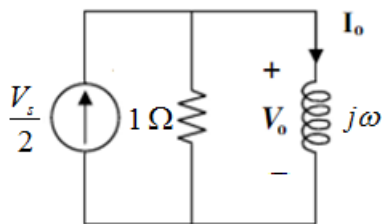


Figure 2

Step 3 of 4

From Figure 2, the output current is,

$$\begin{aligned} \mathbf{I}_o &= \frac{\frac{V_s}{2}}{R + j\omega L} \\ &= \frac{V_s}{2(1 + j\omega)} \end{aligned}$$

The output voltage is,

$$\begin{aligned} V_o &= j\omega I_o \\ &= \frac{j\omega V_s}{2(1 + j\omega)} \end{aligned}$$

Step 4 of 4

Refer to Figure 18.46 (a) in the textbook.

From Figure 18.46(a), the voltage signal is,

$$\begin{aligned} v_s &= 15\delta(t) - 15\delta(t - 2) \\ j\omega V_s(\omega) &= 15 - 15e^{-j2\omega} \\ V_s(\omega) &= \frac{15 - 15e^{-j2\omega}}{j\omega} \\ &= \frac{15(1 - e^{-j2\omega})}{j\omega} \end{aligned}$$

Therefore, the output voltage is

$$\begin{aligned} V_o &= \frac{j\omega \frac{15(1 - e^{-j2\omega})}{j\omega}}{2(1 + j\omega)} \\ &= \frac{15(1 - e^{-j2\omega})}{2(1 + j\omega)} \\ &= \frac{7.5}{(1 + j\omega)} - \frac{7.5e^{-j2\omega}}{(1 + j\omega)} \end{aligned}$$

Apply inverse Fourier transform.

$$v_o(t) = 7.5e^{-t}u(t) - 7.5e^{-(t-2)}u(t-2)$$

At $t = 1\text{ s}$,

$$\begin{aligned} v_o(1) &= 7.5e^{-1}u(1) - 7.5e^{-(1-2)}u(1-2) \\ &= 7.5e^{-1} - 0 \\ &= (7.5)(0.368) \\ &= 2.76 \text{ V} \end{aligned}$$

Thus, the output voltage at $t = 1 \text{ s}$ is 2.76 V.